



PTT PHENOL COMPANY
LIMITED

PROCESS SPECIFICATION

PRESSURE RELIEF VALVE PROCESS DATASHEET

Docu. No. : 14780-8120-PS-0018
Project No. : 120117
Rev. No. : Z1
Date : MAY 10, 2016
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TITLE :

PRESSURE RELIEF VALVE PROCESS DATASHEET

(CDN UNIT)

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DOCUMENT NO. : 14780-8120-PS-0018
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PROJECT NO. : 120117
OWNER : PTT Phenol Company Limited (PPCL)

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PROJECT SPECIFICATION

963766

- 807

SHEET 3/18

PRESSURE RELIEF VALVES

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	10-May-12	MYW	MEH				
1	08-Aug-12	MYW	MEH				

Rev

**UOP Phenol Process Unit
Concentration, Decomposition, and Neutralization Section**

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Other specifications referenced in this specification:	UOP Standard Specification:
	UOP Standard Drawing:
	UOP Project Specification:

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PROJECT SPECIFICATION

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PRESSURE RELIEF VALVES

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	10-May-12	MYW	MEH				

General Notes

- Each pressure relief valve (PRV) shall have identification markings containing all data required by the applicable ASME Code and / or API Standard 526, latest editions.
- All raised face flanges and corresponding mating surfaces shall have a surface finish of 125 microinch Ra minimum to 250 microinch Ra maximum, except as noted. Finishes shall be judged by visual comparison with surface finish roughness standards conforming to ASME B46.1.
- Bolting shall conform to the following ASTM Specifications:
 - Spring-loaded: A193 Grade B7 for bonnet studs and A194 Grade 2H for stud nuts.
 - Pilot-operated: A193 Grade B7 for main valve cap studs and A194 Grade 2H for stud nuts.
- The specification for each valve shows a value for the maximum allowed total back pressure at the valve outlet. This is labeled "Maximum Total Back Pressure". Provided the pressure loss in the relief valve discharge system plus superimposed back pressure does not exceed the "Maximum Total Back Pressure", the valve size specified is valid.
- The Design Temperature and Design Pressure shown on each PRV specification sheet should be used for the design of the pressure-retaining parts of the PRV (i.e. to establish the inlet flange class). The relieving temperatures shown may sometimes be higher than the design temperature. It is UOP's opinion that short-term temperature excursions occurring under relieving cases can be tolerated and need not, in general, impact the design temperature of the PRV or process equipment downstream of the PRV location.
- The Operating Temperature shown on each PRV specification sheet should be used for the design of any temperature-sensitive materials, e.g. elastomeric components, used in the PRV construction. It is UOP's opinion that short-term temperature excursions occurring under relieving cases take time to develop and can generally be tolerated by such components. In the worst case, UOP believes the PRV will open safely, but may not fully close after a high-temperature relief incident, and would therefore need to be taken out of service and have its soft goods replaced to return it to leak-tight operation. Vendor to alert the Customer if in any particular application these assumptions are not valid.
- Equipment items that can be blocked-in require overpressure protection using a PRV or administrative controls to prevent overpressure due to fire, thermal expansion, or blocked outlet. If there is a PRV shown "By Others" at an equipment location on UOP's P&ID, it is the responsibility of the client and/or contractor to determine the method of overpressure protection, and to determine the relief loads based upon purchased equipment and size any PRV's found necessary.
- For a fractionator, the relief rate listed for the external fire case includes the estimated contribution from any heat exchangers associated with that fractionator (but excluding air-cooled condensers or coolers, which are assumed to be above the effective fire height). Once exchanger locations and dimensions are known, the contractor shall confirm the total fire case relief load, and re-size the fractionator PRV as necessary.
- The relief device specifications are to be regarded as preliminary. The contractor shall provide owner, if requested, a relief device documentation package reflecting the final design information including, but not limited to, equipment dimensions and capacities, plot locations, and additional relief cases identified by the contractor.



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PROJECT SPECIFICATION

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SHEET 5/18

PRESSURE RELIEF VALVES

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	10-May-12	MYW	MEH				

Pilot Operated Valve

Rev

General Data							
Item Number(s)		PSV-23-0401A/B/C/D		Design Pressure		3.500	kg/cm ² [g]
Service		Preflash Column		Design Temperature		250	°C
Valves Required		Online:	3	Spare:	1	Operating Pressure	
Valve Action:		Modulating		Operating Temperature		52	°C
Pilot:		Flowing		Atmospheric Pressure		1.03	kg/cm ² [a]
Design Code / Practice:		ASME VIII / API STD 520		Low Ambient Temperature		15	°C
Relief Conditions		Units	Self-Heating Reaction	-	-	-	-
Max Accumulated Pressure, %			116	-	-	-	-
Accumulated Pressure		kg/cm ² [a]	4.47	-	-	-	-
Max Allowable Inlet Line Loss		kg/cm ²	.0889	-	-	-	-
PRV Inlet Cond	Inlet Relieving Pressure	kg/cm ² [a]	4.47	-	-	-	-
	Inlet Temperature	°C	221	-	-	-	-
	Vapor Mass Rate	kg/h	122899	-	-	-	-
	Molecular Weight		125.2	-	-	-	-
	Compressibility Factor		1.00	-	-	-	-
	k = Cp/Cv		1.03	-	-	-	-
	Liquid Volume Rate	m ³ /h	454	-	-	-	-
Relief Header Cond	Sp Gr (water = 1.000)		.697	-	-	-	-
	Liquid Viscosity	cP	0.128	-	-	-	-
	Temperature	°C	163	-	-	-	-
	Vapor Mass Rate	kg/h	308264	-	-	-	-
	Molecular Weight		123.7	-	-	-	-
	Liquid Volume Rate	m ³ /h	167	-	-	-	-
	Sp Gr (water = 1.000)		.782	-	-	-	-
Kd, Coefficient of Discharge			.960	-	-	-	-
Kb, BP Correction Factor (vap)			-	-	-	-	-
Kw, BP Correction Factor (liq)			-	-	-	-	-
Kv, Viscosity Correction Factor (liq)			-	-	-	-	-
KSH, Steam Superheat Correction			-	-	-	-	-
Required Relief Orifice Area		cm ²	601.62	-	-	-	-
Valve			Material Selection		Main Valve		Pilot Valve
Pset (1st & Spare Valve)		2.100	kg/cm ² [g]	Body		316 SS	316 SS
Pset (Other Valves)		2.205	kg/cm ² [g]	Nozzle		316 SS	316 SS
Diff Set Press (1st / Other)		-	kg/cm ²	Seat		Teflon	PEEK
API Orifice Designation				Seals		Teflon	Teflon
Total Effective Orifice Area		754.06	cm ²	Internals		316 SS	316 SS
Flanged or Screwed		Flanged	-	Tubing & Fittings		316 SS	-
Inlet Size (in), Rating, Facing		8 " CL 150 RF				-	-
Outlet Size (in), Rating, Facing		10" x 10" CL 150 RF				-	-
Discharge Conditions			Accessories		Required		Material
Valve Discharges to		To Atmosphere At Safe Location		Pilot Supply Filter		Required	316 SS
Pilot Discharges to		Main Valve Outlet Flange		Field Test Unit		Required	316 SS
Superimposed Back Pressure		0	kg/cm ² [g]	Back Flow Preventer		Required	316 SS
Built-Up Back Pressure		1.04	kg/cm ²	Pressure Spike Snubber		Not Required	-
Maximum Total Back Pressure		1.045	kg/cm ² [g]	Remote Pressure Sensor		Required	316 SS

Notes

1. Provide ½" (13 mm) low point drain on valve body to prevent accumulation of rain water or steam condensate in PRV outlet.
2. Other non-governing case could be Loss of Vacuum in which steam used in the ejectors passes back to the column. The steam will be condensed at preflash and flash column condenser under normal operation.
3. PRV sizing for Self-Heating Reaction case determined using DIERS methodology. Characterization parameter, omega = 2.222
4. Remote sense line must be free-draining from pilot to sense location.
5. "Series 500" (or equal) pilot-operated PRV is required for service conditions. See next note.
6. PRV protects equipment which operates at a vacuum. Backflow preventer required to keep air and/or other contaminants from entering the process.

Notes continue on next sheet.

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PRESSURE RELIEF VALVES

PROJECT SPECIFICATION

963766

- 807

SHEET 6/18

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	10-May-12	MYW	MEH				

Pilot Operated Valve

Rev

Valve Identification

Item Number(s)	PSV-23-0401A/B/C/D
Service	Preflash Column

Notes continued from Sheet 5.

7. Supplier of pilot-operated PRVs: Anderson, Greenwood & Co. PO Box 944 Stafford Texas 77497 USA Telephone (281) 274 4400
8. PRV soft goods (seat and seals) shall be compatible with hydrocarbon containing aromatics at a concentration of 3 mol% or greater.
9. Integral cast or welding neck inlet and outlet flanges are required. Slip on flanges are not permitted.
10. Dual outlet connections may be required to balance reaction forces.

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PROJECT SPECIFICATION

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SHEET 7/18

PRESSURE RELIEF VALVES

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	08-Aug-12	MYW	MEH				

Spring Loaded Valve

Rev

General Data							
Item Number(s)		PSV-23-0501		Design Pressure		7.000 kg/cm ² [g]	
Service		Preflash Column Steam Heater Condensate Drum		Design Temperature		195 °C	
Valves Required		Online: 1 Spare: 0		Operating Pressure		1.251 kg/cm ² [g]	
-		-		Operating Temperature		117 °C	
Type of Valve:		Conventional		Atmospheric Pressure		1.03 kg/cm ² [a]	
Design Code / Practice:		ASME VIII / API STD 520		Low Ambient Temperature		15 °C	
Relief Conditions		Units	External Fire	-	-	-	-
Max Accumulated Pressure, %			121	-	-	-	-
Accumulated Pressure		kg/cm ² [a]	9.50	-	-	-	-
Max Allowable Inlet Line Loss		kg/cm ²	.210	-	-	-	-
PRV Inlet Cond	Inlet Relieving Pressure	kg/cm ² [a]	9.50	-	-	-	-
	Inlet Temperature	°C	177	-	-	-	-
	Vapor Mass Rate	kg/h	157	-	-	-	-
	Molecular Weight		18.0	-	-	-	-
	Compressibility Factor		.933	-	-	-	-
	k = Cp/Cv		1.22	-	-	-	-
	Liquid Volume Rate	m ³ /h	-	-	-	-	-
Relief Header Cond	Sp Gr (water = 1.000)		-	-	-	-	-
	Liquid Viscosity	cP	-	-	-	-	-
	Temperature	°C	-	-	-	-	-
	Vapor Mass Rate	kg/h	-	-	-	-	-
	Molecular Weight		-	-	-	-	-
	Liquid Volume Rate	m ³ /h	-	-	-	-	-
	Sp Gr (water = 1.000)		-	-	-	-	-
Kd, Coefficient of Discharge			.975	-	-	-	-
Kb, BP Correction Factor (vap)			1.00	-	-	-	-
Kw, BP Correction Factor (liq)			-	-	-	-	-
Kv, Viscosity Correction Factor (liq)			-	-	-	-	-
KSH, Steam Superheat Correction			-	-	-	-	-
Required Relief Orifice Area		cm ²	0.32	-	-	-	-
Valve			Material Selection				
Pset (1st & Spare Valve)		7.000 kg/cm ² [g]	Body and Bonnet		Carbon Steel		-
Pset (Other Valves)		- kg/cm ² [g]	Nozzle and Disk		316 SS		-
Diff Set Press (1st / Other)		- kg/cm ²	Spindle and Guide		416 SS / SS		-
API Orifice Designation		D	Spring		Carbon Steel		-
Total Effective Orifice Area		0.71 cm ²	Bellows		-		-
Flanged or Screwed		Flanged	Resilient Seat		-		-
Inlet Size (in), Rating, Facing		1 CL 150 RF	-		-		-
Outlet Size (in), Rating, Facing		2 CL 150 RF	-		-		-
Discharge Conditions			Other Requirements				
Valve Discharges to		Atmosphere	Bonnet Type		Closed		-
Pilot Discharges to		-	Cap		Screwed		-
Superimposed Back Pressure		0 kg/cm ² [g]	Lever		Not Required		-
Built-Up Back Pressure		0.70 kg/cm ²	-		-		-
Maximum Total Back Pressure		0.700 kg/cm ² [g]	-		-		-

Notes

- Relief rate listed for External Fire includes vapor rates from following equipment: D-2308.
- Provide ½" (13 mm) low point drain on valve body to prevent accumulation of rain water or steam condensate in PRV outlet.

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SHEET 8/18

PRESSURE RELIEF VALVES

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	08-Aug-12	MYW	MEH				

Spring Loaded Valve

Rev

General Data							
Item Number(s)	PSV-23-0701			Design Pressure		7.000	kg/cm ² [g]
Service	Flash Column Vaporizer Condensate Drum			Design Temperature		195	°C
Valves Required	Online:	1	Spare: 0	Operating Pressure		2.461	kg/cm ² [g]
-				Operating Temperature		135	°C
Type of Valve:	Conventional		-	Atmospheric Pressure		1.03	kg/cm ² [a]
Design Code / Practice: ASME VIII / API STD 520				Low Ambient Temperature		15	°C
Relief Conditions		Units	External Fire	-	-	-	-
Max Accumulated Pressure, %			121	-	-	-	-
Accumulated Pressure		kg/cm ² [a]	9.50	-	-	-	-
Max Allowable Inlet Line Loss		kg/cm ²	.210	-	-	-	-
PRV Inlet Cond	Inlet Relieving Pressure	kg/cm ² [a]	9.50	-	-	-	-
	Inlet Temperature	°C	177	-	-	-	-
	Vapor Mass Rate	kg/h	157	-	-	-	-
	Molecular Weight		18.0	-	-	-	-
	Compressibility Factor		.933	-	-	-	-
	k = Cp/Cv		1.22	-	-	-	-
	Liquid Volume Rate	m ³ /h	-	-	-	-	-
	Sp Gr (water = 1.000)		-	-	-	-	-
Relief Header Cond	Liquid Viscosity	cP	-	-	-	-	-
	Temperature	°C	-	-	-	-	-
	Vapor Mass Rate	kg/h	-	-	-	-	-
	Molecular Weight		-	-	-	-	-
	Liquid Volume Rate	m ³ /h	-	-	-	-	-
Sp Gr (water = 1.000)			-	-	-	-	-
Kd, Coefficient of Discharge			.975	-	-	-	-
Kb, BP Correction Factor (vap)			1.00	-	-	-	-
Kw, BP Correction Factor (liq)			-	-	-	-	-
Kv, Viscosity Correction Factor (liq)			-	-	-	-	-
KSH, Steam Superheat Correction			-	-	-	-	-
Required Relief Orifice Area		cm ²	0.32	-	-	-	-
Valve			Material Selection				
Pset (1st & Spare Valve)		7.000	kg/cm ² [g]	Body and Bonnet		Carbon Steel	-
Pset (Other Valves)		-	kg/cm ² [g]	Nozzle and Disk		316 SS	-
Diff Set Press (1st / Other)		-	- kg/cm ²	Spindle and Guide		416 SS / SS	-
API Orifice Designation		D		Spring		Carbon Steel	-
Total Effective Orifice Area		0.71	cm ²	Bellows		-	-
Flanged or Screwed		Flanged	-	Resilient Seat		-	-
Inlet Size (in), Rating, Facing		1	CL 150 RF	-		-	-
Outlet Size (in), Rating, Facing		2	CL 150 RF	-		-	-
Discharge Conditions				Other Requirements			
Valve Discharges to		Atmosphere		Bonnet Type		Closed	-
Pilot Discharges to		-		Cap		Screwed	-
Superimposed Back Pressure		0	kg/cm ² [g]	Lever		Not Required	-
Built-Up Back Pressure		0.70	kg/cm ²	-		-	-
Maximum Total Back Pressure		0.700	kg/cm ² [g]	-		-	-

Notes

- Relief rate listed for External Fire includes vapor rates from following equipment: D-2309.
- Provide ½" (13 mm) low point drain on valve body to prevent accumulation of rain water or steam condensate in PRV outlet.



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SHEET 9/18

PRESSURE RELIEF VALVES

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	10-May-12	MYW	MEH				

Pilot Operated Valve

Rev

General Data							
Item Number(s)		PSV-23-0801A/B/C/D/E		Design Pressure		3.500	kg/cm ² [g]
Service		Flash Column		Design Temperature		250	°C
Valves Required		Online:	4	Spare:	1	Operating Pressure	
Valve Action:		Modulating		Operating Temperature		52	°C
Pilot:		-	Flowing	Atmospheric Pressure		1.03	kg/cm ² [a]
Design Code / Practice:		ASME VIII / API STD 520		Low Ambient Temperature		15	°C
Relief Conditions		Units	Self-Heating Reaction	-	-	-	-
Max Accumulated Pressure, %			116	-	-	-	-
Accumulated Pressure		kg/cm ² [a]	3.89	-	-	-	-
Max Allowable Inlet Line Loss		kg/cm ²	.0738	-	-	-	-
PRV Inlet Cond	Inlet Relieving Pressure	kg/cm ² [a]	3.89	-	-	-	-
	Inlet Temperature	°C	213	-	-	-	-
	Vapor Mass Rate	kg/h	215684	-	-	-	-
	Molecular Weight		130.3	-	-	-	-
	Compressibility Factor		1.00	-	-	-	-
	k = Cp/Cv		1.03	-	-	-	-
	Liquid Volume Rate	m ³ /h	252	-	-	-	-
Relief Header Cond	Sp Gr (water = 1.000)		.728	-	-	-	-
	Liquid Viscosity	cP	0.122	-	-	-	-
	Temperature	°C	168	-	-	-	-
	Vapor Mass Rate	kg/h	327745	-	-	-	-
	Molecular Weight		128.7	-	-	-	-
	Liquid Volume Rate	m ³ /h	89.0	-	-	-	-
	Sp Gr (water = 1.000)		.804	-	-	-	-
Kd, Coefficient of Discharge			.970	-	-	-	-
Kb, BP Correction Factor (vap)			-	-	-	-	-
Kw, BP Correction Factor (liq)			-	-	-	-	-
Kv, Viscosity Correction Factor (liq)			-	-	-	-	-
KSH, Steam Superheat Correction			-	-	-	-	-
Required Relief Orifice Area		cm ²	718.03	-	-	-	-
Valve			Material Selection		Main Valve		Pilot Valve
Pset (1st & Spare Valve)		2.100	kg/cm ² [g]	Body		316 SS	316 SS
Pset (Other Valves)		2.205	kg/cm ² [g]	Nozzle		316 SS	316 SS
Diff Set Press (1st / Other)		-	kg/cm ²	Seat		Teflon	PEEK
API Orifice Designation				Seals		Teflon	Teflon
Total Effective Orifice Area		1005.42	cm ²	Internals		316 SS	316 SS
Flanged or Screwed		Flanged	-	Tubing & Fittings		316 SS	-
Inlet Size (in), Rating, Facing		8 " CL 150 RF				-	-
Outlet Size (in), Rating, Facing		10" x 10" CL 150 RF				-	-
Discharge Conditions			Accessories		Required		Material
Valve Discharges to		To Atmosphere At Safe Location		Pilot Supply Filter		Required	316 SS
Pilot Discharges to		Main Valve Outlet Flange		Field Test Unit		Required	316 SS
Superimposed Back Pressure		0	kg/cm ² [g]	Back Flow Preventer		Required	316 SS
Built-Up Back Pressure		0.985	kg/cm ²	Pressure Spike Snubber		Not Required	-
Maximum Total Back Pressure		0.9846	kg/cm ² [g]	Remote Pressure Sensor		Required	316 SS

Notes

- Provide ½" (13 mm) low point drain on valve body to prevent accumulation of rain water or steam condensate in PRV outlet.
- Other non-governing case could be Loss of Vacuum in which steam used in the ejectors passes back to the column. The steam will be condensed at preflash and flash column condenser under normal operation.
- PRV sizing for Self-Heating Reaction case determined using DIERS methodology. Characterization parameter, omega = 1.436
- Remote sense line must be free-draining from pilot to sense location.
- "Series 500" (or equal) pilot-operated PRV is required for service conditions. See next note.
- PRV protects equipment which operates at a vacuum. Backflow preventer required to keep air and/or other contaminants from entering the process.

Notes continue on next sheet.

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PRESSURE RELIEF VALVES

PROJECT SPECIFICATION

963766

- 807

SHEET 10/18

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	10-May-12	MYW	MEH				

Pilot Operated Valve

Rev

Valve Identification

Item Number(s)	PSV-23-0801A/B/C/D/E
Service	Flash Column

Notes continued from Sheet 9.

7. Supplier of pilot-operated PRVs: Anderson, Greenwood & Co. PO Box 944 Stafford Texas 77497 USA Telephone (281) 274 4400
8. PRV soft goods (seat and seals) shall be compatible with hydrocarbon containing aromatics at a concentration of 3 mol% or greater.
9. Integral cast or welding neck inlet and outlet flanges are required. Slip on flanges are not permitted.
10. Dual outlet connections may be required to balance reaction forces.

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PROJECT SPECIFICATION

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SHEET 11/18

PRESSURE RELIEF VALVES

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	10-May-12	MYW	MEH				

Spring Loaded Valve

Rev

General Data							
Item Number(s)		PSV-23-1403A/B/C		Design Pressure		21.00	kg/cm ² [g]
Service		Dehydrator Reactor Loop X-2312		Design Temperature		250	°C
Valves Required		Online:	3	Spare:	0	Operating Pressure	
-		-		Operating Temperature		139	°C
Type of Valve:		Balanced Bellows		Atmospheric Pressure		1.03	kg/cm ² [a]
Design Code / Practice:		ASME VIII / API STD 520		Low Ambient Temperature		15	°C
Relief Conditions		Units	Blocked In - Thermal Expansion	-	-	-	-
Max Accumulated Pressure, %			110	-	-	-	-
Accumulated Pressure		kg/cm ² [a]	23.6	-	-	-	-
Max Allowable Inlet Line Loss		kg/cm ²	0.62	-	-	-	-
PRV Inlet Cond	Inlet Relieving Pressure	kg/cm ² [a]	23.6	-	-	-	-
	Inlet Temperature	°C	139	-	-	-	-
	Vapor Mass Rate	kg/h	-	-	-	-	-
	Molecular Weight		-	-	-	-	-
	Compressibility Factor		-	-	-	-	-
	k = Cp/Cv		-	-	-	-	-
	Liquid Volume Rate	m ³ /h	.00273	-	-	-	-
Relief Header Cond	Sp Gr (water = 1.000)		.832	-	-	-	-
	Liquid Viscosity	cP	0.305	-	-	-	-
	Temperature	°C	-	-	-	-	-
	Vapor Mass Rate	kg/h	-	-	-	-	-
	Molecular Weight		-	-	-	-	-
	Liquid Volume Rate	m ³ /h	-	-	-	-	-
	Sp Gr (water = 1.000)		-	-	-	-	-
Kd, Coefficient of Discharge			.650	-	-	-	-
Kb, BP Correction Factor (vap)			-	-	-	-	-
Kw, BP Correction Factor (liq)			0.88	-	-	-	-
Kv, Viscosity Correction Factor (liq)			.799	-	-	-	-
KSH, Steam Superheat Correction			-	-	-	-	-
Required Relief Orifice Area		cm ²	0.00	-	-	-	-
Valve			Material Selection				
Pset (1st & Spare Valve)		20.50	kg/cm ² [g]	Body and Bonnet		316 SS	-
Pset (Other Valves)		-	kg/cm ² [g]	Nozzle and Disk		316 SS	-
Diff Set Press (1st / Other)		-	kg/cm ²	Spindle and Guide		416 SS / SS	-
API Orifice Designation		D		Spring		Alloy Steel	-
Total Effective Orifice Area		0.71	cm ²	Bellows		Inconel	-
Flanged or Screwed		Flanged	-	Resilient Seat		-	-
Inlet Size (in), Rating, Facing		1	CL 300 RF	-		-	-
Outlet Size (in), Rating, Facing		2	CL 150 RF	-		-	-
Discharge Conditions			Other Requirements				
Valve Discharges to		Acid Aromatics Knockout Drum		Bonnet Type		Closed	-
Pilot Discharges to		-		Cap		Screwed	-
Superimposed Back Pressure		0.50	kg/cm ² [g]	Lever		Not Required	-
Built-Up Back Pressure		5.65	kg/cm ²	Liquid Trim		Required	-
Maximum Total Back Pressure		6.150	kg/cm ² [g]	-		-	-

Notes

1. On-line PRVs are in identical or parallel service. Each PRV is sized for 100% of relief rate.
2. Valve requires liquid capacity certification.
3. Set pressure lower than design pressure due to liquid static head.

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PROJECT SPECIFICATION

963766

- 807

SHEET 12/18

PRESSURE RELIEF VALVES

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	10-May-12	MYW	MEH				

Spring Loaded Valve

Rev

General Data							
Item Number(s) PSV-23-1801A/B and PSV-23-1802A/B				Design Pressure		10.50	kg/cm ² [g]
Service Calorimeters				Design Temperature		250	°C
Valves Required Online: 2 Spare: 2				Operating Pressure		2.109	kg/cm ² [g]
-				Operating Temperature		79	°C
Type of Valve: Balanced Bellows				Atmospheric Pressure		1.03	kg/cm ² [a]
Design Code / Practice: ASME VIII / API STD 520				Low Ambient Temperature		15	°C
Relief Conditions		Units	Blocked In - Thermal Expansion	Blocked Outlet	-	-	-
Max Accumulated Pressure, %			110	110	-	-	-
Accumulated Pressure		kg/cm ² [a]	11.5	11.5	-	-	-
Max Allowable Inlet Line Loss		kg/cm ²	0.29	0.29	-	-	-
PRV Inlet Cond	Inlet Relieving Pressure	kg/cm ² [a]	11.5	11.5	-	-	-
	Inlet Temperature	°C	79	79	-	-	-
	Vapor Mass Rate	kg/h	-	-	-	-	-
	Molecular Weight		-	-	-	-	-
	Compressibility Factor		-	-	-	-	-
	k = Cp/Cv		-	-	-	-	-
	Liquid Volume Rate	m ³ /h	.000253	.152	-	-	-
	Sp Gr (water = 1.000)		.898	.898	-	-	-
Relief Header Cond	Liquid Viscosity	cP	0.570	0.571	-	-	-
	Temperature	°C	-	-	-	-	-
	Vapor Mass Rate	kg/h	-	-	-	-	-
	Molecular Weight		-	-	-	-	-
	Liquid Volume Rate	m ³ /h	-	-	-	-	-
	Sp Gr (water = 1.000)		-	-	-	-	-
Kd, Coefficient of Discharge			.650	.650	-	-	-
Kb, BP Correction Factor (vap)			-	-	-	-	-
Kw, BP Correction Factor (liq)			0.88	0.88	-	-	-
Kv, Viscosity Correction Factor (liq)			.1000	.970	-	-	-
KSH, Steam Superheat Correction			-	-	-	-	-
Required Relief Orifice Area		cm ²	0.00	0.02	-	-	-
Valve				Material Selection			
Pset (1st & Spare Valve)		9.50	kg/cm ² [g]	Body and Bonnet		316 SS	-
Pset (Other Valves)		-	kg/cm ² [g]	Nozzle and Disk		316 SS	-
Diff Set Press (1st / Other)		-	kg/cm ²	Spindle and Guide		416 SS / SS	-
API Orifice Designation		D		Spring		Alloy Steel	-
Total Effective Orifice Area		0.71	cm ²	Bellows		Inconel	-
Flanged or Screwed		Flanged	-	Resilient Seat		-	-
Inlet Size (in), Rating, Facing		1 CL 150 RF		-		-	-
Outlet Size (in), Rating, Facing		2 CL 150 RF		-		-	-
Discharge Conditions				Other Requirements			
Valve Discharges to		Acid Aromatics Knockout Drum		Bonnet Type		Closed	-
Pilot Discharges to		-		Cap		Screwed	-
Superimposed Back Pressure		0.50	kg/cm ² [g]	Lever		Not Required	-
Built-Up Back Pressure		2.35	kg/cm ²	Liquid Trim		Required	-
Maximum Total Back Pressure		2.850	kg/cm ² [g]	-		-	-

Notes

- On-line PRVs are in identical or parallel service. Each PRV is sized for 100% of relief rate.
- Valve requires liquid capacity certification.
- Set pressure lower than design pressure due to liquid static head.

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PROJECT SPECIFICATION

963766

- 807

SHEET 13/18

PRESSURE RELIEF VALVES

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	08-Aug-12	MYW	MEH				

Spring Loaded Valve

Rev

General Data											
Item Number(s)	PSV-23-2001A/B			Design Pressure		3.500	kg/cm ² [g]				
Service	Acid Aromatic Sump			Design Temperature		325	°C				
Valves Required	Online:	1	Spare:	1	Operating Pressure		.007031	kg/cm ² [g]			
-				-		Operating Temperature		68	°C		
Type of Valve:				Balanced Bellows		-		Atmospheric Pressure		1.03	kg/cm ² [a]
Design Code / Practice:				ASME VIII / API STD 520		Low Ambient Temperature		15		°C	
Relief Conditions		Units	External Fire	-	-	-	-	-			
Max Accumulated Pressure, %			121	-	-	-	-	-			
Accumulated Pressure		kg/cm ² [a]	5.27	-	-	-	-	-			
Max Allowable Inlet Line Loss		kg/cm ²	.105	-	-	-	-	-			
PRV Inlet Cond	Inlet Relieving Pressure	kg/cm ² [a]	5.27	-	-	-	-	-			
	Inlet Temperature	°C	179	-	-	-	-	-			
	Vapor Mass Rate	kg/h	6687	-	-	-	-	-			
	Molecular Weight		62.5	-	-	-	-	-			
	Compressibility Factor		.941	-	-	-	-	-			
	k = Cp/Cv		1.08	-	-	-	-	-			
	Liquid Volume Rate	m ³ /h	-	-	-	-	-	-			
Sp Gr (water = 1.000)			-	-	-	-	-	-			
Relief Header Cond	Liquid Viscosity	cP	-	-	-	-	-	-			
	Temperature	°C	-	-	-	-	-	-			
	Vapor Mass Rate	kg/h	-	-	-	-	-	-			
	Molecular Weight		-	-	-	-	-	-			
	Liquid Volume Rate	m ³ /h	-	-	-	-	-	-			
Sp Gr (water = 1.000)			-	-	-	-	-	-			
Kd, Coefficient of Discharge			.975	-	-	-	-	-			
Kb, BP Correction Factor (vap)			1.0	-	-	-	-	-			
Kw, BP Correction Factor (liq)			-	-	-	-	-	-			
Kv, Viscosity Correction Factor (liq)			-	-	-	-	-	-			
KSH, Steam Superheat Correction			-	-	-	-	-	-			
Required Relief Orifice Area		cm ²	14.33	-	-	-	-	-			
Valve			Material Selection								
Pset (1st & Spare Valve)		3.500	kg/cm ² [g]	Body and Bonnet		Carbon Steel		-			
Pset (Other Valves)		-	kg/cm ² [g]	Nozzle and Disk		316 SS		-			
Diff Set Press (1st / Other)		-	-	kg/cm ²		Spindle and Guide		416 SS / SS	-		
API Orifice Designation		L		Spring		Alloy Steel		-			
Total Effective Orifice Area		18.41	cm ²	Bellows		Inconel		-			
Flanged or Screwed		Flanged		-		Resilient Seat		-			
Inlet Size (in), Rating, Facing		3	CL 150	RF	-	-		-			
Outlet Size (in), Rating, Facing		4	CL 150	RF	-	-		-			
Discharge Conditions				Other Requirements							
Valve Discharges to		Relief Header		Bonnet Type		Closed		-			
Pilot Discharges to		-		Cap		Screwed		-			
Superimposed Back Pressure		0.50	kg/cm ² [g]	Lever		Not Required		-			
Built-Up Back Pressure		0.55	kg/cm ²	-		-		-			
Maximum Total Back Pressure		1.050	kg/cm ² [g]	-		-		-			

Notes

- Back pressure at valve outlet affects valve capacity.



PROJECT SPECIFICATION

963766 - 808 SHEET 14/18

RUPTURE DISCS

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	09May12	WHC	MYW				
1	1July14	MYW	WHC				

**UOP Phenol Process Unit
Concentration, Decomposition, and Neutralization Section**

- Item Number :	X-2311
Location :	Top of Decomposer Drum D-2304

- Rupture Disc	
Quantity	One (1) operating plus two (2) spares
Size	24" (Note 1)
Type	B (Note 2)
Pressure and Temperature	
Normal Operating Conditions	0.7 kg/cm2(g) at 60°C
Burst Pressure at Coincident Temperature (Note 3)	MAWP of D-2304 at 60°C Not to exceed 11.0 kg/cm2(g) at 250°C Not to be less than 9.5 kg/cm2(g) at 225°C
Manufacturing Range	See Burst Pressure (above)
Disc Material	316 Stainless Steel
Liner Material, process side	Not required
Liner Material, downstream (atmospheric) side	Teflon or equal

- Safety Head	
Quantity	One (1) operating plus two (2) spares
Size	24" (Note 1)
Type	No. FA-7R (Note 2) safety head including preassembly screws for installation between two ASME Class 300 raised face pipe flanges
Inlet Material	316 Stainless Steel
Outlet Material	316 Stainless Steel

- Accessories	
Jack Screws	Required

- Note 1: 24 inch disc provided for pressure relief in unlikely event that safety instrumented system fails.
- Note 2: BS&B Safety Systems nomenclature.
- Note 3: Estimated Burst Pressure at all indicated temperatures to be reported by supplier following design of rupture disc.

Other specifications referenced in this specification:	UOP Standard Specification:
	UOP Standard Drawing:
	UOP Project Specification:

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RUPTURE DISCS

PROJECT SPECIFICATION

963766 - 808 SHEET 15/18

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	09-May-12	WHC	MYW				

Notes

1. Fluid Handled

Safety head assembly is to be installed on top of the Decomposer Drum (D-2304). It is used to protect the vessel from rupture during a highly unlikely, but possible uncontrolled reaction. Material containing nitrogen, acetone and phenol will be discharged to the atmosphere at a safe location.

2. Safety Head Flange mating surfaces

The mating surfaces of the safety head with the companion flanges shall have a surface finish of 125 Ra minimum to 250 Ra maximum. Surface finishes shall be judged by visual comparison with surface finish roughness standards conforming to ASME B46.1. Finishes shall be protected from damage during shipping, storage, and installation.

3. Bolting

Supplied by others (Bolting to span two welding neck flanges, two gaskets, and safety head assembly). Safety head supplier shall provide allowable bolting torque information, if applicable.

4. Gaskets

Supplied by others (Gaskets to be spiral wound, low-stress type, e.g. Flexitallic, with graphite filler).

5. Accessories

Provide a pipe cap for an angle-cut discharge of a 24-inch pipe, with durable cable tether. Contractor shall determine suitable anchoring location on the vent pipe for the tether.

Supplier shall provide installation instructions.

6. Code Compliance

Disk shall be stamped in accordance with ASME Boiler and Pressure Vessel Code (if required by user).

ASME capacity certification is not required

Supplier (or Equal)

- BS&B Safety Systems
- www.bsbsystems.com

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Other specifications referenced in this specification:	UOP Standard Specification:
	UOP Standard Drawing:
	UOP Project Specification:

						PRESSURE RELIEF VALVES PROCESS SPECIFICATION			
CUSTOMERS NAME: PTT PHENOL COMPANY LIMITED						PROJECT No: 120117			
LOCATION: MAP TA PHUT, THAILAND						UNIT No: 2300			
						DOCUMENT No. 14780-8120-PS-0018			
REV	A1	F1	F2	Z1	PAGE No. 16 / 18				
DATE	MAR 26, 2014	JUN 3, 2014	JAN 13, 2015	MAY 10, 2016					
ORIG. BY	J.K.JUNG	J.K.JUNG	J.K.JUNG	J.K.JUNG					
APP. BY	S.W.PARK	S.W.PARK	S.W.PARK	S.W.PARK					
Item No.		PSV-23-0301A/B		PSV-23-0802		PSV-23-0803		PSV-23-1001	
P&ID No.		14780-8120-25-23-0003		14780-8120-25-23-0008		14780-8120-25-23-0008A		14780-8120-25-23-0010	
Service		X-2302A/B PROTECTION		E-2306 TUBE SIDE PROTECTION		E-2301 PLATE SIDE PROTECTION		E-2310 TUBE SIDE PROTECTION	
Fluid Description		HYDROCARBON		RELIABLE CW		RELIABLE CW		CHILLED WATER	
State		VAP		LIQ		LIQ		LIQ	
Corrosive Compounds		N/A		N/A		N/A		N/A	
Number of valves required		2		1		1		1	
In regular use		1		1		1		1	
As spares		1		N/A		N/A		N/A	
Interlocks required		NO		NO		NO		NO	
Governing conditions		EXTERNAL FIRE		THERMAL EXPANSION		THERMAL EXPANSION		THERMAL EXPANSION	
Relief Valve type		CONVENTIONAL		CONVENTIONAL		CONVENTIONAL		CONVENTIONAL	
Minimum orifice area req'd (4) mm ²		194.9		MIN.		MIN.		MIN.	
Relief Valve size (4)		1-1/2G3		3/4" x 1"		3/4" x 1"		3/4" x 1"	
Flow per valve at relieving conditions kg/h		2945		136		2981		58	
Compressibility factor		1.00		-		-		-	
Vapour Cp/Cv		1.03		-		-		-	
Molecular weight		134.90		18.02		18.02		21.00	
S.G. at relieving conditions		0.68		0.9989		0.9989		1.048	
Viscosity Liquid/Vapour cP		/ 0.012		0.612 /		0.612 /		2.063 /	
Liquid vapour press at relieving temp. kg/cm ² (a)		-		-		-		-	
Operating temperature min/nor/max °C		/ 82 /		/ 33 /		/ 33 /		/ 0 /	
Relieving temperature °C		250.8		43.0		43.0		10.0	
Operating pressure min/nor/max kg/cm ² g		/ 5.5 /		/ 6.4 /		/ 6.4 /		/ 6.8 /	
Set pressure kg/cm ² g		11.0 (Note 5)		10.5		10.5		11.5	
Superimposed back pressure (constant) kg/cm ² g		0.0100		0.0100		0.0100		0.0100	
Superimposed back pressure (variable)(1) kg/cm ² g		- TO -		- TO -		- TO -		- TO -	
Built-up back pressure(2) kg/cm ² g		1.09		1.04		1.04		1.14	
Maximum Total Back pressure (3) kg/cm ² g		1.10		1.05		1.05		1.15	
Allowable accumulation %		21		10		10		10	
Flashing Service: wt% vapour at outlet		-		-		-		-	
Relief to		DECANTER (ATM)		ATM		ATM		ATM	
Date		MAY 10, 2016		MAY 10, 2016		MAY 10, 2016		MAY 10, 2016	
Revision									
Remarks									
NOTES 1. Specify full range. 2. This is a differential pressure which must be added to the Superimposed back pressure to give the Back pressure at the relief valve outlet. 3. Maximum Back pressure is the sum of Superimposed and Built up back pressures. 4. Preliminary only. To be confirmed by Instrument Group. 5. Set pressure has been adjusted due to liquid static head.									

					PRESSURE RELIEF VALVES PROCESS SPECIFICATION			
CUSTOMERS NAME: PTT PHENOL COMPANY LIMITED					PROJECT No: 120117			
LOCATION: MAP TA PHUT, THAILAND					UNIT No: 2300			
					DOCUMENT No. 14780-8120-PS-0018			
REV	A1	F1	F2	Z1	PAGE No. 17 / 18			
DATE	MAR 26, 2014	JUN 3, 2014	JAN 13, 2015	MAY 10, 2016				
ORIG. BY	J.K.JUNG	J.K.JUNG	J.K.JUNG	J.K.JUNG				
APP. BY	S.W.PARK	S.W.PARK	S.W.PARK	S.W.PARK				

Item No.	PSV-23-1401A/B		PSV-23-1404	PSV-23-1406	PSV-23-1407
P&ID No.	14780-8120-25-23-0014		14780-8120-25-23-0014A	14780-81250-25-23-0014	14780-81250-25-23-0014
Service	E-2308A/B TUBE SIDE PROTECTION		E-2309 SHELL SIDE PROTECTION	E-2308A SHELL SIDE PROTECTION	E-2308B SHELL SIDE PROTECTION
Fluid Description	HYDROCARBON		COOLING WATER	STEAM	STEAM
State	LIQ		VAP	VAP	VAP
Corrosive Compounds	N/A		N/A	N/A	N/A
Number of valves required	2		1	1	1
In regular use	1		1	1	1
As spares	1		N/A	N/A	0
Interlocks required	NO		NO	NO	NO
Governing conditions	THERMAL EXPANSION		EXTERNAL FIRE	EXTERNAL FIRE	EXTERNAL FIRE
Relief Valve type	CONVENTIONAL		CONVENTIONAL	CONVENTIONAL	CONVENTIONAL
Minimum orifice area req'd (4)	mm ²	MIN.	79.5	70.1	70.1
Relief Valve size (4)	3/4" x 1"		1E2	1D2	1D2
Flow per valve at relieving conditions	kg/h	4012	862	761	761
Compressibility factor	-		1.00	1.00	1.00
Vapour Cp/Cv	-		1.48	1.70	1.70
Molecular weight	77.30		18.02	18.02	18.02
S.G. at relieving conditions	0.79		-	-	-
Viscosity Liquid/Vapour	cP	0.2995 /	/ 0.0170	/ 0.0160	/ 0.0160
Liquid vapour press at relieving temp.	kg/cm ² (a)	-	-	-	-
Operating temperature min/nor/max	°C	/ 83 /	/ 33 /	/ 148 /	/ 148 /
Relieving temperature	°C	145	214	214	214.0
Operating pressure min/nor/max	kg/cm ² g	/ 5.6 /	/ 3.9 /	/ 2.7 /	/ 2.7 /
Set pressure	kg/cm ² g	19.8 (Note 5)	16.5	16.5	16.5
Superimposed back pressure (constant)	kg/cm ² g	0.5000	0.0100	0.0100	0.0100
Superimposed back pressure (variable)(1)	kg/cm ² g	- TO -	- TO -	- TO -	- TO -
Built-up back pressure(2)	kg/cm ² g	1.48	1.64	1.64	1.64
Maximum Total Back pressure (3)	kg/cm ² g	1.98	1.65	1.65	1.65
Allowable accumulation	%	10	21	21	21
Flashing Service: wt% vapour at outlet				-	-
Relief to	HP FLARE		ATM	ATM	ATM
Date	MAY 10, 2016		MAY 10, 2016	MAY 10, 2016	MAY 10, 2016
Revision					
Remarks					

NOTES

- Specify full range.
- This is a differential pressure which must be added to the Superimposed back pressure to give the Back pressure at the relief valve outlet.
- Maximum Back pressure is the sum of Superimposed and Built up back pressures.
- Preliminary only. To be confirmed by Instrument Group.
- Set pressure has been adjusted due to liquid static head.

					PRESSURE RELIEF VALVES PROCESS SPECIFICATION					
CUSTOMERS NAME: PTT PHENOL COMPANY LIMITED					PROJECT No: 120117					
LOCATION: MAP TA PHUT, THAILAND					UNIT No: 2300					
					DOCUMENT No. 14780-8120-PS-0018					
REV	A1	F1	F2	Z1	PAGE No. 18 / 18					
DATE	MAR 26, 2014	JUN 3, 2014	JAN 13, 2015	MAY 10, 2016						
ORIG. BY	J.K.JUNG	J.K.JUNG	J.K.JUNG	J.K.JUNG						
APP. BY	S.W.PARK	S.W.PARK	S.W.PARK	S.W.PARK						
Item No.					PSV-23-1701		PSV-23-1405A/B		PSV-23-1903A/B	
P&ID No.					14780-8120-25-23-0017		14780-8120-25-23-0014A		14780-8120-25-23-0019	
Service					E-2307A/B SHELL SIDE PROTECTION		E-2309 TUBE SIDE PROTECTION		D-2312 PROTECTION	
Fluid Description					RELIABLE CW		HYDROCARBON		DIAMINE	
State					VAP		VAP		VAP	
Corrosive Compounds					N/A		N/A		N/A	
Number of valves required					1		2		2	
In regular use					1		1		1	
As spares					0		1		1	
Interlocks required					NO		YES		YES	
Governing conditions					EXTERNAL FIRE		EXTERNAL FIRE		EXTERNAL FIRE	
Relief Valve type					CONVENTIONAL		CONVENTIONAL		BELLOWS	
Minimum orifice area req'd (4) mm ²					283.0		29.7		338.5	
Relief Valve size (4)					1-1/2G3		1D2		1-1/2H3	
Flow per valve at relieving conditions kg/h					2479		835		1012	
Compressibility factor					1.00		1.00		0.92	
Vapour Cp/Cv					1.46		1.32		1.33	
Molecular weight					18.02		78.30		21.91	
S.G. at relieving conditions					-		-			
Viscosity Liquid/Vapour cP					/ 0.0163		/ 0.0130		/ 0.0120 /	
Liquid vapour press at relieving temp. kg/cm ² (a)					-					
Operating temperature min/nor/max °C					/ 33 /		/ 43 /		/ 40 / /	
Relieving temperature °C					202.8		225.0		301.0	
Operating pressure min/nor/max kg/cm ² g					/ 6.4 /		/ 5.5 /		/ 0.0 / /	
Set pressure kg/cm ² g					13.0		20.5 (Note 5)		3.5	
Superimposed back pressure (constant) kg/cm ² g					0.0100		0.5000		0.5000	
Superimposed back pressure (variable)(1) kg/cm ² g					- TO -		- TO -		- TO -	
Built-up back pressure(2) kg/cm ² g					1.29		1.55		1.25	
Maximum Total Back pressure (3) kg/cm ² g					1.30		2.05		1.75	
Allowable accumulation %					21		21		21	
Flashing Service: wt% vapour at outlet									-	
Relief to					ATM		HP FLARE		HP FLARE	
Date					MAY 10, 2016		MAY 10, 2016		MAY 10, 2016	
Revision										
Remarks										
NOTES 1. Specify full range. 2. This is a differential pressure which must be added to the Superimposed back pressure to give the Back pressure at the relief valve outlet. 3. Maximum Back pressure is the sum of Superimposed and Built up back pressures. 4. Preliminary only. To be confirmed by Instrument Group. 5. Set pressure has been adjusted due to liquid static head.										